# Technical Design Document

Cubli

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# 1 Introduction

## 1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype (15 × 15 × 15 cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

**Why this is a hard build, not a decorative demo.** It is worth stating plainly why a machine that looks like a cube with three flywheels resists casual replication. The difficulty is not that any single subsystem is exotic — none is — but that every subsystem sits close to a physical bound, and the bounds are *coupled*: relieving one tightens another. The following failure modes recur across published and amateur replications, and each is a consequence of that coupling rather than of poor workmanship.

- **The jump-up is an energy problem with no margin.** The wheel must store, at its maximum speed, enough angular momentum to lift the cube's center of mass over the corner. The required per-wheel inertia scales as $ML^{3/2}$ (Section 4.3.1), so mass added *anywhere* — battery, frame, fasteners — demands a larger wheel, and the wheel is itself part of $M$. The sizing problem is therefore a fixed-point iteration rather than a one-pass calculation, and it does not always converge: for the present cube the three-wheel set alone accounts for roughly a quarter of the total mass budget, and shrinking the wheel diameter to save space makes the situation worse, not better, because inertia must then be recovered with ballast at a smaller radius.
- **Braking torque fights the structure.** The momentum accumulated over seconds of spin-up must be shed in tens of milliseconds. That impulse passes through one motor shaft, one hub, and one motor mount — the very parts the mass budget wants thin. Many builds fail here first, and they fail in a characteristic way: the electronics are fine, the controller is sound, and the coupling, hub or mount yields on the first *successful* jump. The structural load case is set by the maneuver that proves the machine works.

- **Sensing, not the control law, sets the balancing limit.** The attitude estimate degrades from two directions at once: MEMS gyroscope bias drifts, and the accelerometer — the only absolute reference for tilt — is corrupted by vibration that the controller itself generates by spinning the wheels. Structural design and state estimation are consequently not separable concerns. A wheel half a gram out of balance at rim radius can render a perfectly correct estimator unusable, which is why wheel balance is treated here as an estimation requirement and not merely as a mechanical nicety.

- **The plant is genuinely nonlinear and constrained.** Corner balancing does not decouple into three independent single-axis problems: the axes are kinematically coupled through the contact point and the body inertia tensor. The actuators saturate in *both* torque and speed, and under any persistent disturbance the wheels drift toward their speed limit, so stored momentum must be actively managed rather than left to accumulate. Linearizing about the upright equilibrium is entirely adequate for a balancing demonstration and inadequate for anything more interesting [3].

The common thread is worth making explicit, because it determines how this project is sequenced. Replications typically fail not because the control law was wrong, but because the machine was never capable of the maneuver being asked of it: insufficient wheel inertia, motors that cannot reach their rated speed under load, a brake too slow to approximate an impulse, or a mass budget that quietly grew past the point where the stored energy suffices. Every one of these is decidable on paper, before any material is cut — and commonly is not decided there. That is the reason sizing is treated in this document as the first engineering activity rather than a detail to be settled once parts arrive.

## 1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli's three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. Reaction wheels are distinct from control moment gyros (CMGs), which gimbal a constantly-spinning wheel instead of changing its speed to produce much larger torques [4]; positioning the Cubli explicitly as a *reaction-wheel*, not CMG, demonstrator keeps the application claim precise.

Framed as an ADCS problem, the Cubli's core behaviors map directly onto a spacecraft's day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.

- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.

- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible. JAXA's Hayabusa2 mission deployed the MINERVA-II1 rovers [5] and the MASCOT lander [6] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH's SpaceHopper [7]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control* (ADCS), and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers *"where am I pointing?"* (attitude determination), the Cubli answers *"drive me to that reference and hold it"* (attitude control) — the two halves of a complete ADCS loop.

**Why the problem is still open.** The nominal Cubli problem — balance on a corner, jump up — is solved, and was solved a decade ago on well-instrumented, precisely manufactured, generously funded hardware [1, 2, 3]. Reproducing that result is an engineering exercise, not a research contribution, and this document does not claim otherwise. The interesting questions begin immediately *after* that point, and they are not Cubli-specific: they are the questions a spacecraft attitude-control engineer faces, in a form small enough to be answered on a bench.

- **Control under simultaneous torque and momentum saturation.** A jump saturates the wheels by construction — the maneuver is defined by running them to their speed limit and dumping the momentum. Steering a nonlinear attitude system to a target while respecting a finite momentum envelope, and desaturating without losing the pointing objective, is still handled largely case-by-case with tuned heuristics rather than by a general method.
- **Recovery from large attitude excursions.** Linear controllers valid near upright say nothing useful once the body is far from equilibrium, which is precisely the regime a tumbling or mis-deployed spacecraft occupies. Nonlinear and predictive methods make claims there, and verifying those claims empirically requires a system in which a failed recovery costs a re-print rather than a mission.
- **Robustness to badly known parameters.** Real inertia tensors, friction coefficients and center-of-mass offsets are known to a few percent at best, and shift with assembly, thermal state and wear. Adaptive and robust designs are overwhelmingly validated against *simulated* parameter errors, which are drawn from the same distribution the designer assumed.
- **Whether high-performance methods survive commodity sensing.** Attitude-control techniques developed against clean measurements must eventually run on a MEMS IMU with drifting bias and vibration-corrupted accelerometers. Whether a given method degrades gracefully or collapses under that sensing is an empirical question, and it is settled by hardware rather than by analysis.

**Why a Cubli, and why built from scrap.** The Cubli is the cheapest honest test of the four questions above. Simulation cannot answer them, because the effects that decide the outcome are exactly the ones a model omits: unmodeled friction, frame resonance excited by the wheels, and the interaction between wheel imbalance and estimator bias. These are what separate a controller that works in simulation from one that works on hardware. At the other extreme, a full-scale testbed — air bearing, vacuum chamber, flight-grade wheels — costs orders of magnitude more and, decisively, *cannot be crashed repeatedly.* The scrap-built cube occupies the useful middle: physical enough to be honest, cheap enough to be abused. The choice to build it from commodity and salvaged parts is therefore methodological rather than merely budgetary, on four counts:

- **Cheap failure means the aggressive experiments actually get run.** A controller that must never fail cannot be tested at its limits, and a testbed too valuable to break silently biases the experimental program toward maneuvers already known to work.

- **Cheap parts are a stronger test than good parts.** Loose tolerances, an imperfectly balanced wheel and a drifting IMU are not defects to be apologized for — they *are* the parameter uncertainty and disturbance content that robust and adaptive methods claim to tolerate. A demonstration on deliberately imperfect hardware therefore carries evidence that a demonstration on precision hardware does not.
- **Constraints force honest sizing.** With a fixed motor, a fixed print volume and a fixed budget, a marginal design cannot be rescued by specifying a larger component. This is the entire justification for the bolt-in-pocket ballast scheme of Section 5.3.4: it buys inertia at the outer radius using standard M6 steel bolts and no machined part, and whether it closes is a calculation that must be completed *before* printing, not discovered afterwards.
- **A commodity bill of materials is reproducible.** Every part is orderable or salvageable by any university group, so the result can be checked by others — which is the difference between a demonstration and a claim.

**Mission context.** The subsystem exercised here is the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. In one respect the terrestrial case is strictly *harder* than the orbital one. In orbit, a marginal attitude controller degrades slowly and often invisibly — pointing error grows, wheels creep toward saturation, and the fault may take orbits to become obvious. Under gravity, the destabilizing torque never vanishes and never relents: an inadequate controller falls over immediately and unambiguously. The demonstration is, in that sense, self-evaluating, and this is a feature of the platform rather than a limitation of it.

### 1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

# 2 Project Organization

## 2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member's primary role, and the detailed responsibilities are outlined below.

**Table 1:** Team members, contact details and primary roles.

| **Member** | **Contact** | **Primary Role** |
|---|---|---|
| Pablo Urioste Alarcón | pb.urioste4@gmail.com<br>+34 654 491 174 | System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design. |
| Deyan Nikolaev Vlaev | vlaev.dido@gmail.com<br>+359 877 082 499 | Mathematical demonstration and derivation of the system dynamics. |
| Niccolo | — | Control system design and analysis. |
| Andrea Liang | andrea.liang.2006@gmail.com<br>+31 06 5796 2976 | Support on the control problem and structural/mechanical integration. |
| Suvanna Viriyanti Wu | Viriyanti.wu@gmail.com<br>+39 329 074 7550 | CAD modeling and mechanical design. |
| Athanasia Nikolova | — | CAD modeling, prototype building/assembly, and product presentation (commercial pitch). |

**Responsibilities.**

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

## 2.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The documentation stream is front-loaded and closes with the Technical Design Document deliverable on 3 August, while the control and hardware streams continue through the build, balancing and jump-up demonstrations up to the presentation. The work is grouped into three streams — investigation, requirements and documentation; control and software; and hardware design and build — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. The table below breaks the three streams into individual tasks, each with its scope, owner and dates.

| ID | Task | Scope | Owner | Dates |
|---|---|---|---|---|
| **Investigation, Requirements & Documentation** (to the 3 Aug TDD deliverable) | | | | |
| A1 | Literature & state of the art | Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents | Suvanna *(+Dejan)* | 23–25 Jul |
| A2 | Problem statement & framing | Lock the ADCS application narrative that the rest of the document refers back to | Dejan | 23–24 Jul |
| A3 | Scope & requirements | Define scope and build the requirements table (functional, performance, design constraint) with verification methods | Dejan *(+Pablo)* | 24–25 Jul |
| A4 | Milestones & roles | Prototype staging, binary gate criteria, and a one-page RACI of owners | Dejan | 23–25 Jul |
| A5 | Risk register | Trigger, impact, owner and mitigation for the leading risks | All *(+Pablo)* | 28–29 Jul |
| A6 | 1-DoF equations of motion | Lagrangian derivation of the reaction-wheel pendulum with friction terms | Dejan *(+Andrea)* | 24–25 Jul |
| A7 | 3D equations of motion | Coupled three-wheel dynamics for the edge and corner pivots | Dejan *(+Andrea)* | 27–28 Jul |
| A8 | Linearisation & controllability | Equilibria, linear models, controllability/observability checks | Dejan *(+Andrea)* | 29 Jul |
| A9 | Jump-up sizing & budget | Energy-method wheel sizing and the parametric mass/inertia/CoM budget | Dejan *(+Andrea)* | 30 Jul–1 Aug |
| A10 | TDD draft (all sections) | Every member drafts their section; placeholder figures acceptable | All | 25–26 Jul |
| A11 | TDD deepen + S1 data | Full modelling and control sections; insert the rig evidence and measured parameters | All | 1–2 Aug |
| A12 | TDD assemble & submit | Consistency pass, figures, references, submission of the deliverable | All *(Dejan integrates)* | 3 Aug |
| **Control & Software** | | | | |
| B1 | Sim environment + plant | Simulation setup and the 1-DoF plant, validated by energy conservation | Andrea | 23–24 Jul |
| B2 | LQR design (1-DoF, sim) | Tune the regulator and sweep robustness to inertia error, delay and saturation | Andrea *(+Dejan)* | 26–27 Jul |
| B3 | State estimator | Complementary tilt filter with injected noise and bias in simulation | Nicc | 27–28 Jul |
| B4 | Firmware & comms | Firmware architecture, loop rates, and the CAN-FD command/reply layer | Nicc *(+Pablo)* | 24–28 Jul |
| B5 | Sensor drivers + cal | IMU and encoder drivers, gyro/accel calibration and the lever-arm correction | Nicc *(+Andrea)* | 28–30 Jul |
| B6 | Safety & mode logic | Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine | Nicc *(+Andrea)* | 30–31 Jul |

| ID | Task | Scope | Owner | Dates |
|---|---|---|---|---|
| B7 | S1 bring-up & balance | Port estimator and LQR to the rig and tune to a stable balance | Nicc *(+Andrea)* | 31 Jul–2 Aug |
| B8 | System identification | Measure $K_t$, wheel inertia, friction and loop latency on the rig | Andrea *(+Nicc)* | 1–2 Aug |
| B9 | 3D model + LQR (sim) | 3D plant and the edge/corner regulators with the CAD inertia tensor | Andrea *(+Dejan)* | 3–6 Aug |
| B10 | EKF / MEKF estimator | 3D estimator with gyro-bias states; characterise the yaw drift | Nicc *(+Andrea)* | 6–8 Aug |
| B11 | Edge balance (S2a) | Retune the edge controller on the assembled cube; log disturbance recovery | Andrea *(+Nicc)* | 10–13 Aug |
| B12 | Corner balance + slew (S2b/c) | Coupled three-axis corner balance and a commanded slew; report ADCS metrics | Andrea *(+Nicc)* | 14–16 Aug |
| B13 | Jump-up (S3) | Spin-up profile and brake release for a face-to-edge jump | Nicc *(+Andrea)* | 17–18 Aug |
| **Hardware Design & Build** | | | | |
| C1 | Envelope & layout study | Packaging of wheels, drivers, battery and electronics with a central CoM | Neisa *(+Suvanna)* | 23–24 Jul |
| C2 | Reaction-wheel design | Wheel to the inertia target: radius, rim, hub bolting to the motor bell | Suvanna *(+Neisa)* | 24–26 Jul |
| C3 | Mounts & brackets | Motor mount, ring-magnet carrier and adjustable encoder bracket | Neisa, Suvanna | 27–29 Jul |
| C4 | 1-DoF rig CAD + jigs | Single-axis rig and the balancing/assembly tooling | Suvanna *(+Neisa)* | 25–28 Jul |
| C5 | Mass properties & freeze | Export the inertia tensor/CoM and release the manufacturing package | Neisa *(+Suvanna)* | 2–3 Aug |
| C6 | Procurement order | Order the long-lead items (rims, frame stock, connectors, spares) | Pablo *(+Neisa)* | 23–24 Jul |
| C7 | Bench PSU & bring-up | Current-limited supply, safety kit, and single-axis electrical bring-up | Pablo *(+Nicc)* | 25–27 Jul |
| C8 | Fabricate wheels (×4) | Cut/receive rims, print hubs, bond magnets — three plus one rig spare | Neisa, Suvanna | 27–30 Jul |
| C9 | Balance & spin-test | Static-balance and containment spin-test every wheel | Suvanna *(+Neisa)* | 30–31 Jul |
| C10 | Assemble 1-DoF rig | Motor, balanced wheel, encoder gap, IMU and hard stops | Pablo *(+Neisa)* | 31 Jul–1 Aug |
| C11 | Print structure + frame | Structural print set and the fabricated, orthogonal frame | Neisa, Suvanna | 3–6 Aug |
| C12 | Perfboard + harness | Power/signal board and the full power and CAN harness | Pablo | 3–6 Aug |
| C13 | Cube integration + power-on | Assemble the three axes, then bench and battery smoke test | Pablo *(+Neisa)* | 7–9 Aug |
| C14 | Brake mechanism build | Fabricate and fit the servo brake; bench-test the deceleration | Suvanna *(+Pablo)* | 10–13 Aug |
| **Electrical & Electronics** (electronics design: Nicc; wiring, testing and bring-up: Pablo) | | | | |
| E1 | Bench power, e-stop & triage | Current-limited bench supply, e-stop and inline fuse; incoming test of all three actuation sets against the 194 mΩ phase figure, with CAN IDs assigned | Pablo | 28–29 Jul |

| ID | Task | Scope | Owner | Dates |
|---|---|---|---|---|
| E2 | Driver, phase wiring & 5 V rail | Motor phase wiring and FOC calibration of the first set; LM2596 rail trimmed to 5.00 V and characterised under load to 1.5 A | Pablo *(+Nicc)* | 29–30 Jul |
| E3 | MA600 encoder + rig CAN bus | Magnet bonding, air gap $\leq 0.5$ mm and bias trim; 2-node CAN-FD bus with split termination at both physical ends | Pablo | 30–31 Jul |
| E4 | Rig parameters + board outline | Measurement setup for $K_t$, wheel inertia, friction and loop latency; driver placement, CAN routing and the perfboard floorplan issued as a frozen outline | Nicc *(+Pablo)* | 31 Jul–1 Aug |
| E5 | 1-DoF closed-loop bring-up | Teensy, IMU and CAN-FD closed loop on the rig; disarm path and watchdog verified before any balance attempt | Pablo *(+Nicc)* | 1–2 Aug |
| E6 | Perfboard + battery power chain | Populate the distribution perfboard (two ground domains, one star point, split 5 V rail); XT90 chain with anti-spark, fuse and $\geq 50$ V bulk capacitance; replicate the remaining actuation sets | Nicc *(+Pablo)* | 3–4 Aug |
| E7 | 4-node bus + cube harness | Bus scaled to four nodes with stubs $< 10$ cm and an error soak at 5 Mbit/s; full power and signal harness fabricated and routed; Wi-Fi telemetry for untethered operation | Pablo | 5–7 Aug |
| E8 | Cube power-on, 48 A + thermal | First battery power-on and the three-axis 48 A transient; driver thermals in the enclosed volume; all axes addressable | Pablo *(+Nicc)* | 7–9 Aug |
| E9 | Brake servo rail + measurement | Servo on an isolated 5 V rail with PWM drive and flyback; rail stability under stall, brake closure time and spin-down brake torque measured | Pablo *(+Nicc)* | 10–18 Aug |
| **Deliverables** | | | | |
| D1 | Results, demo & final presentation | Slow-motion capture, results plots, deck, and the final presentation | All *(+Suvanna edits)* | 17–20 Aug |

## 2.3 Schedule and Contingency Plan

The full-page schedule in Figure 1 places every task of Section 2.2 on the 23 July–20 August 2026 calendar, and Table 3 states the gate that closes each phase. The documentation stream is deliberately completed early, at the TDD deliverable on 3 August, so the remaining weeks are free for the build, balancing and jump-up work that culminates in the final presentation. The critical path runs through the dynamics investigation and wheel sizing, which feed both the control and the hardware streams; the single-axis (S1) rig is the protected item, since its balance result is the experimental evidence in the document. If any modeling or rig task slips, the TDD is still submitted on schedule with simulation-only evidence and the affected result carried into the later build weeks.

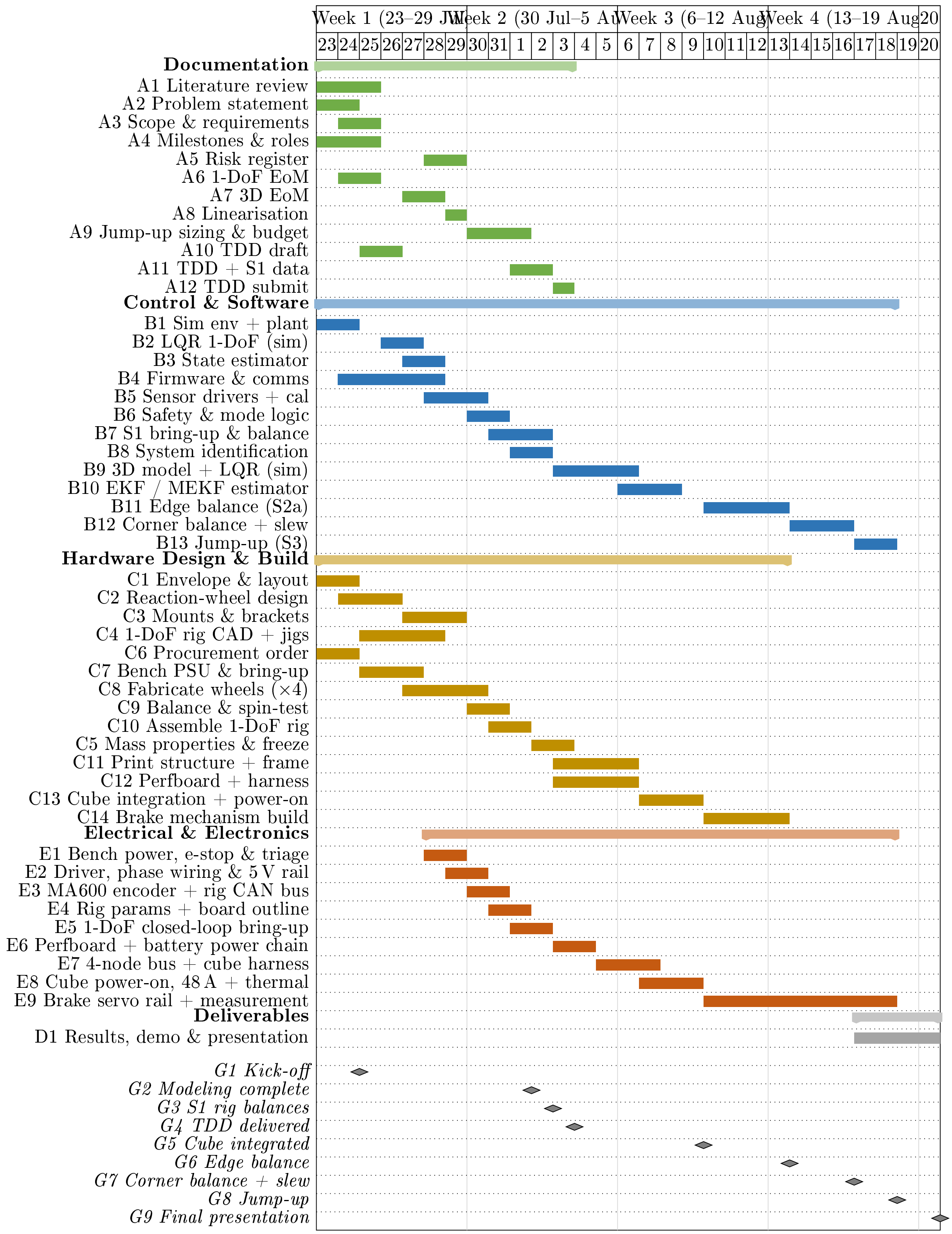


**Figure 1:** Detailed project schedule, 23 July–20 August 2026. The documentation stream (green) closes at the TDD deliverable on 3 August; control (blue), hardware (gold) and electrical (orange) continue through the build and demonstrations to the final presentation on 20 August. The electrical stream covers the design, wiring and testing of the 1-DoF rig (E1–E5, to G3), its extension to the 3D cube (E6–E8, to G5) and the brake actuation electrical (E9, to G8); each bar summarises several tasks of the detailed electrical schedule. Grey diamonds: milestone gates.

**Table 3:** Milestone gates, from kick-off to the final presentation.

| Gate | Pass criterion | Date |
|---|---|---|
| G1 | Long-lead order placed; requirements and plan locked | 24 Jul |
| G2 | Dynamics, control design and wheel sizing complete | 1 Aug |
| G3 | 1-DoF rig balances $\geq$60 s; plant parameters measured | 2 Aug |
| G4 | TDD delivered (documentation deliverable) | 3 Aug |
| G5 | Cube integrated and powered; all axes addressable | 9 Aug |
| G6 | Edge balance (S2a) for $\geq$60 s with disturbance recovery | 13 Aug |
| G7 | Corner balance and commanded slew (S2b/c) | 16 Aug |
| G8 | Face-to-edge jump-up (S3) captured on video | 18 Aug |
| G9 | Final presentation and demonstration | 20 Aug |

# 3 System Analysis & Requirements

## 3.1 Scope of Work & Use Cases

## 3.2 Actuation Trade-off — Why Reaction Wheels

## 3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

**Table 4:** Objective-to-requirement-to-verification traceability.

| Objective | Required capability | Verified by |
|---|---|---|
| Edge balance | Regulation about a line-contact equilibrium | Sec. 6 |
| Corner balance | Regulation about a point-contact equilibrium, all axes | Sec. 6 |
| Controlled rotation | Reference tracking / slew between equilibria | Sec. 6 |
| Walking (stretch) | Chained jump-up + re-stabilization | Sec. 6 |

## 3.4 Coordinate Frames & Conventions

# 4 Dynamic Modeling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modeling, identification, and control workflow is therefore first validated on a two-dimensional (2D) single-face prototype with one rotational degree of freedom, before being generalized to the 3D cube in Section 4.5.

## 4.1 2D Reaction-Wheel Pendulum

The 2D prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its center and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

### 4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l)\, g \sin\theta_b - T_m - C_b\dot{\theta}_b + C_w\dot{\theta}_w}{I_b + m_w l^2}, \tag{1}$$

$$\begin{aligned}\ddot{\theta}_w = &\frac{(I_b + I_w + m_w l^2)(T_m - C_w\dot{\theta}_w)}{I_w(I_b + m_w l^2)} \\ &- \frac{(m_b l_b + m_w l)\, g \sin\theta_b - C_b\dot{\theta}_b}{I_b + m_w l^2},\end{aligned} \tag{2}$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

**Table 5:** 2D pendulum model variables.

| Symbol | Description | Unit |
|---|---|---|
| $\theta_b$ | Body tilt angle | rad |
| $\theta_w$ | Wheel rotation relative to body | rad |
| $m_b, m_w$ | Body, wheel mass | kg |
| $I_b, I_w$ | Body (about pivot), wheel (about motor axis) inertia | $\mathrm{kg\,m^2}$ |
| $l$ | Motor axis to pivot distance | m |
| $l_b$ | Body CoM to pivot distance | m |
| $g$ | Gravitational acceleration | $\mathrm{m/s^2}$ |
| $T_m$ | Motor torque | N m |
| $C_b, C_w$ | Body, wheel damping coefficient | $\mathrm{kg\,m^2/s}$ |
| $K_m$ | Motor torque constant | N m/A |
| $u$ | Motor current input | A |

### 4.1.2 Equilibria & Linearization

The relevant equilibrium is the upright position $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$; its linearization is presented with the control design in Section 4.4.

## 4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives $m_b, m_w, I_b, I_w$ directly, while $l$ and $l_b$ follow from the plate's design geometry. The damping coefficients $C_b$ and $C_w$ cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

**Table 6:** Parameter determination method (symbols per Table 5).

| Symbol | Determination method |
|---|---|
| $l$ | Calculated during design |
| $l_b$ | Calculated during design |
| $m_b$ | CAD mass properties |
| $m_w$ | CAD mass properties |
| $I_b$ | CAD mass properties |
| $I_w$ | CAD mass properties |
| $C_b$ | Estimated (future testing planned with prototype) |
| $C_w$ | Estimated (future testing planned with prototype) |
| $K_m$ | Motor datasheet |

## 4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed $\omega_w$ required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2 - \sqrt{2})\, \frac{I_w + I_b + m_w l^2}{I_w^2}\, (m_b l_b + m_w l)\, g. \tag{3}$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, $\omega_w$ is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the "instantaneous stop" assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

### 4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\mathrm{ideal}} = \sqrt{\eta}\, \frac{\sqrt{2g\,\lambda\,\kappa}}{n}\, M L^{3/2}, \tag{4}$$

where $M, L, g$ are cube mass, edge length, and gravity, $\eta$ is an energy margin, and $\kappa, \lambda, n$ are dimensionless geometric factors set by the contact case (Table 7).

The corner case is more demanding and governs the design.

**Table 7:** Contact-case geometric factors.

| Case | $\kappa$ (pivot inertia) | $\lambda$ (CoM rise) | $n$ (momentum transfer) |
|---|---|---|---|
| Edge | $2/3$ | $1/\sqrt{2} - 1/2$ | $1$ |
| Corner (governs design) | $11/12$ | $\sqrt{3}/2 - 1/2$ | $\sqrt{2}$ |

A finite brake torque $\tau_b$ cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \qquad h_w = \beta\, h_{w,\text{ideal}}, \qquad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \tag{5}$$

where $\omega_{\max}$ is the maximum practical wheel speed. Equations (4) and (5) are the sizing relations used at step 4 of the closure procedure of Section 5.2: given a candidate cube mass and edge length they return the inertia the wheel of Section 5.3.4 must meet. Their inputs are recorded in Table 8 and are not fixed here—$M$ and $L$ are outputs of the packaging closure, not assumptions preceding it.

Two features of these relations govern how the iteration behaves. First, $h_w \propto ML^{3/2}$, so growing the cube to accommodate a larger wheel also raises the requirement; convergence depends on the achievable inertia growing faster ($\sim D_w^2$) than the requirement does. Second, $\beta$ diverges as $\tau_b \to \tau_g$: a brake whose torque only marginally exceeds the gravitational tip-over torque cannot deliver the required impulse at any wheel speed. The brake torque must therefore be checked against $\tau_g = MgL/2$ on every iteration, and a comfortable ratio $\tau_b/\tau_g$ maintained rather than a merely positive margin.

**Table 8:** Torque-limited wheel-inertia target (corner case). Inputs are fixed by the closure procedure of Section 5.2; the table is populated on each iteration.

| Symbol | Description | Value |
|---|---|---|
| $M$ | Cube mass | TBD |
| $L$ | Cube edge length | TBD |
| $\eta$ | Energy margin | TBD |
| $\tau_b$ | Brake torque per wheel | TBD |
| $\tau_g = MgL/2$ | Gravitational tip-over floor | TBD |
| $\beta$ | Brake-amplification factor | TBD |
| $\omega_{\max}$ | Max wheel speed (design) | TBD |
| $h_w$ | Required angular momentum | TBD |
| $I_{w,\text{target}}$ | **Target wheel inertia** | **TBD** |

**Choice of $\omega_{\max}$.** The maximum wheel speed is a design choice rather than a datasheet figure, and it enters the sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$: a higher assumed speed buys a proportionally smaller wheel. It is therefore the natural relief valve when the envelope caps the wheel below the requirement (step 5, Table 11), but it must be set from what the drive can sustain in service, not from the no-load figure $K_v V_{\text{bus}}$. Three effects separate the two: winding resistance, which costs speed under the torque required to spin the wheel up; the reduced phase voltage available under sinusoidal field-oriented modulation; and the fall of pack voltage through the discharge curve, which must be taken at its lower end if the maneuver is to remain available on a partly-depleted pack rather than only on a fresh one. The design value is set below the ceiling those effects imply, leaving margin for motor tolerance, bearing and aerodynamic drag, and the residual uncertainty in $M$. The resulting back-EMF should remain a comfortable fraction of the usable bus voltage, and the electrical frequency $\omega_{\max} p/2\pi$ at $p$ pole pairs must sit inside the driver's commutation capability—both checks to be confirmed once the motor and pack are selected.

## 4.4 Control Design on the 2D Model

### 4.4.1 LQR Baseline Control

Linearizing about the upright equilibrium $(\theta_b, \dot\theta_b, \dot\theta_w) = (0, 0, 0)$ gives

$$\dot x(t) = Ax(t) + Bu(t), \qquad x := (\theta_b, \dot\theta_b, \dot\theta_w), \tag{6}$$

with $A, B$ built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_d x[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimizes

$$J(u) = \sum_k x^{\mathrm{T}}[k]Qx[k] + u^{\mathrm{T}}[k]Ru[k], \tag{7}$$

giving $u[k] = -K_d(\hat\theta_b[k], \hat{\dot\theta}_b[k], \hat{\dot\theta}_w[k])$.

The tilt estimate $\hat\theta_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's center. This placement is necessary because an accelerometer at radius $r$ from the pivot measures gravity plus the tangential ($r\ddot\theta_b$) and centripetal ($r\dot\theta_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat\theta_b[k] = (1-\gamma)\big(\hat\theta_b[k-1] + \dot\theta_{b,\mathrm{gyro}}[k]\,\Delta t\big) + \gamma\,\theta_{b,\mathrm{accel}}[k], \tag{8}$$

where $\theta_{b,\mathrm{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0,1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset $d$, including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1-\alpha)x_f[k] + \alpha\hat\theta_b[k] \tag{9}$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat\theta_b[k] - x_f[k],\, \hat{\dot\theta}_b[k],\, \hat{\dot\theta}_w[k]), \tag{10}$$

so $d$ is absorbed by $x_f$ at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 9 summarizes the control variables.

### 4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat\theta_b| < \theta_{\mathrm{th}}$, with hysteresis around $\theta_{\mathrm{th}}$ to prevent mode chattering. The baseline uses a single fixed gain $K_d$ at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearization weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearization points, with the controller interpolating between them based on $\hat\theta_b$, reusing the same discrete structure and offset filter.

**Table 9:** Control-design variables (2D and corner-balancing 3D model).

| Symbol | Description |
|---|---|
| $x$ | State vector |
| $A, B$ | Continuous-time state/input matrices (linearized model) |
| $A_d, B_d$ | Discrete-time (ZOH) counterparts of $A, B$ |
| $K_d$ | LQR feedback gain |
| $Q, R$ | LQR state/input weighting matrices |
| $J(u)$ | LQR cost function |
| $\hat{\theta}_b, \hat{\dot{\theta}}_b$ | Estimated tilt angle, rate (gyroscope/accelerometer) |
| $\hat{\dot{\theta}}_w$ | Estimated wheel rate (encoder) |
| $a_x, a_y$ | Raw accelerometer readings (tangential, radial axes) |
| $\theta_{b,\text{accel}}$ | Instantaneous accelerometer-only tilt estimate |
| $\dot{\theta}_{b,\text{gyro}}$ | Raw gyro rate measurement |
| $\gamma$ | Complementary-filter weight (accelerometer term) |
| $\Delta t$ | Control/estimation loop sample period |
| $x_f, \alpha$ | Offset-correction filter state, filter coefficient |
| $d$ | Constant tilt-sensor offset |
| $\theta_{\text{th}}$ | Tilt threshold, jump-up $\to$ balancing switch |
| $K_d^{(i)}$ | Scheduled gain at linearization point $i$ (extension) |

## 4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

### 4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R\,[\omega_b]_\times, \tag{11}$$

where $[\cdot]_\times$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^{3} I_{w,i}\,\omega_{w,i}\,e_i. \tag{12}$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b\,\omega_b, \tag{13}$$

and each wheel obeys

$$I_{w,i}\big(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}\big) = T_{m,i} - C_{w,i}\,\omega_{w,i}, \qquad i = 1, 2, 3. \tag{14}$$

Table 10 defines all symbols.

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D prototype model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

**Table 10:** Full-body (3D) model variables.

| Symbol | Description |
|---|---|
| $R$ | Cube orientation relative to world frame |
| $\omega_b$ | Body angular velocity, body frame ($\in \mathbb{R}^3$) |
| $\omega_{b,i} = e_i^{\mathrm{T}} \omega_b$ | Body angular velocity about wheel axis $i$ |
| $\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$ | Wheel angular velocities relative to body |
| $e_i$ | Spin axis of wheel $i$ (face normal) |
| $I$ | $3 \times 3$ inertia tensor, body+wheels, about pivot |
| $H$ | Total angular momentum about pivot |
| $\tau_g = m\, r_{cm} \times (R^{\mathrm{T}} \hat{g})$ | Gravity torque about pivot |
| $m$ | Total assembly mass |
| $r_{cm}$ | Body-frame vector, pivot to assembly CoM |
| $\hat{g}$ | World vertical unit vector |
| $C_b$ | $3 \times 3$ body damping matrix |
| $I_{w,i}, C_{w,i}$ | Inertia, damping of wheel $i$ |
| $T_{m,i} = K_{m,i} u_i$ | Motor torque, wheel $i$ |
| $u_i, K_{m,i}$ | Motor current, torque constant, wheel $i$ |

### 4.5.2 Edge vs. Corner Equilibria & Linearization

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with $m_b, l_b, I_b$ replaced by their edge equivalents.

**Corner balancing** is genuinely 3D: all three wheels contribute across three rotational DOF. Linearizing about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \tag{15}$$

with $A \in \mathbb{R}^{9\times 9}$ and $B \in \mathbb{R}^{9\times 3}$ built from $I$, $r_{cm}$, and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10). The LQR design generalizes directly: $K_d$ becomes a $3\times 9$ gain matrix, using the same discretization, cost function, and offset filter as Table 9, with $Q, R$ weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

# 5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

## 5.1 System Architecture

Figure 2 shows the closed control loop that ties the four subsystems together. The Teensy 4.1 runs estimation and control at 1 kHz and issues torque commands over CAN-FD to three moteus-n1 drivers, each closing a 30 kHz field-oriented current loop on its MN4006 motor. The resulting reaction torque acts on the cube body; the body's attitude is sensed by the BMI270 IMU and the rotor angle by the MA600 absolute encoder, closing the loop back to the controller. The two loops run at different rates by design: the fast current loop is delegated entirely to the drivers, so the flight controller need only meet the 1 ms outer-loop deadline of Section 4.4.

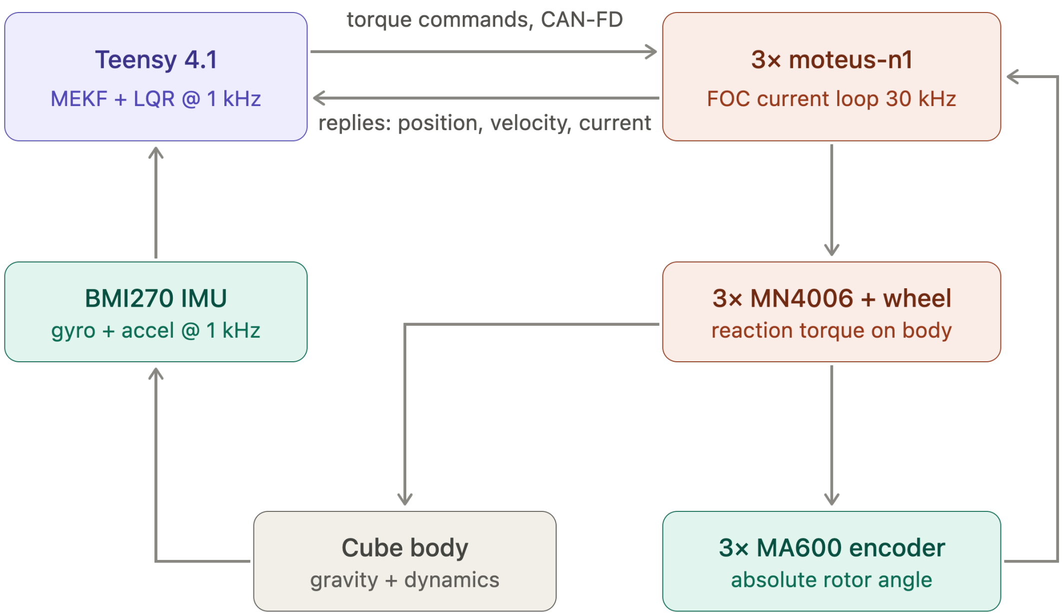


**Figure 2:** System architecture and control loop. Blue: processing; green: sensing; orange: actuation. Rates shown are the design targets of Section 4.

## 5.2 Sizing Constraints and Closure Procedure

The cube edge length $L$ and the reaction-wheel inertia $I_w$ are not independent design choices: they are the two ends of a single constraint chain that begins with what must physically be carried inside the cube. This section states that chain and the order in which it is closed. No dimensions are committed here—component selection is still open (Section 5), and the numbers follow from the procedure rather than preceding it.

**The battery and the internal structure drive the envelope.** The dominant volume claim inside the cube is not the reaction wheels but the power and support hardware. A LiPo pack sized for the jump-up duty cycle is a monolithic, incompressible block: unlike the electronics it cannot be redistributed into the corners or thinned, and its dimensions follow from the required capacity and discharge rating rather than from any packaging preference. Around it must sit an *inner* *structure*—a rigid subframe that locates the three orthogonal motors on mutually perpendicular axes through a common centre, carries the three motor drivers, the flight controller, the IMU and the brake servos, and reacts the motor mounting loads into the outer frame. The three wheel/motor modules must not intersect one another, the pack, or this subframe, and the pack should sit near the geometric centre so that it does not skew the cube's centre of mass away from the body diagonal.

The minimum internal clear volume is therefore set by the pack, the subframe and the electronics stack together, and $L$ is the smallest edge length that contains that volume plus the wall thickness and the corner and edge contact features. This is the first quantity to fix, and everything downstream depends on it.

**The envelope caps the wheel, and the wheel must meet the momentum requirement.** Once $L$ is set, the largest wheel outer diameter $D_w$ that clears the subframe, the motor body and the adjacent orthogonal modules is a geometric consequence, and it caps the achievable inertia—not only through $I_w \sim m_w D_w^2/4$, but because any ballast necessarily sits inboard of the ring it displaces, so a wheel that is short of target at its maximum diameter cannot be rescued by adding mass. Against this stands the requirement from the jump-up analysis (Section 4.3.1),

$$I_{w,\text{target}} = \frac{h_w(M,L)}{\omega_{\max}}, \qquad h_w \propto ML^{3/2},$$

which grows with both the cube mass and the edge length. Enlarging the cube to gain wheel diameter therefore also raises the requirement—but the achievable inertia grows faster than the requirement does ($D_w^2$ against $L^{3/2}$), so the loop does converge, and a wheel that is short at a given $L$ can be closed by a modest increase in $L$.

**Mass closes the loop.** The cube mass $M$ is not merely a logistics figure: it re-enters the dynamics through the gravitational tip-over torque $\tau_g = MgL/2$, which sets the floor the brake torque must exceed, and through $h_w \propto M$. Growing the envelope to fit a larger wheel adds frame and wheel mass, which raises $I_{w,\text{target}}$ again. Mass, volume and inertia are thus one coupled system, closed by the iteration of Table 11 rather than by any of the three in isolation.

**Table 11:** Sizing closure procedure. Steps are executed in order; a failure at step 4 returns to step 3 or step 1 as indicated.

| Step | Determines | Governing consideration |
|---|---|---|
| 1 | Battery pack geometry | Capacity and discharge rating for the jump-up duty cycle |
| 2 | Inner clear volume $V_{\min}$, hence minimum $L$ | Pack + motor subframe + driver/controller stack, non-intersecting; pack near centre |
| 3 | Maximum wheel diameter $D_w$, hence achievable $I_w$ | Clearance to subframe, motor body and the two orthogonal modules |
| 4 | Requirement $I_{w,\text{target}} = h_w(M,L)/\omega_{\max}$ | Corner jump-up, Eqs. (4)–(5) |
| 5 | Accept or iterate | If $I_w < I_{w,\text{target}}$: raise $L$ (step 2), raise $\omega_{\max}$, or trim $M$; re-enter at step 3 |
| 6 | Mass reconciliation | Updated $M$ feeds back into step 4; iterate to a fixed point |

**Inertia requirement is corner-critical.** The two contact cases are not equally demanding. The corner equilibrium requires a larger rise of the centre of mass and a less favourable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia (Table 7). Step 4 above is therefore always evaluated for the corner case; a wheel that satisfies the corner requirement satisfies the edge requirement with margin. Table 12 records both once the procedure closes.

**Table 12:** Required per-wheel inertia by contact case, to be evaluated at the selected $\omega_{\max}$ once the envelope is fixed.

| Contact case | $h_w$ (kg m$^2$/s) | $I_{w,\text{target}}$ (kg m$^2$) |
|---|---|---|
| Edge | TBD | TBD |
| **Corner (critical)** | TBD | TBD |

**Mass budget.** The allocation of Table 13 is the bookkeeping side of steps 5 and 6: it is populated as components are selected and re-totalled on each iteration, since the total is an input to $I_{w,\text{target}}$ and not merely an output of the design.

**Table 13:** Mass budget (per cube), to be populated as components are selected. The total is an input to the inertia requirement, not only a result.

| Subsystem | Allocated mass (g) | Status |
|---|---|---|
| Battery pack | TBD | Drives envelope (step 1) |
| Reaction wheels (3×) | TBD | Pending Sec. 5.3.4 |
| Motors (3×) | TBD | Pending motor selection |
| Inner structure / motor subframe | TBD | Pending CAD |
| Outer frame / contact features | TBD | Pending CAD |
| Electronics (MCU, drivers, IMU, servos) | TBD | Allowance |
| Wiring / fasteners / margin | TBD | Allowance |
| **Total $M$** | **TBD** | **Closed by iteration** |

## 5.3 Mechanical Design and CAD Construction

This subsection states the mechanical concept and the CAD conventions the build follows. It is deliberately kept to the level of decisions and interfaces: the part-by-part model tree, the manufacturing drawings, the fabrication settings and the assembly sequence are recorded in Appendix B, so that a change to a bracket does not require an edit to the design rationale, and the rationale is not buried in geometry.

### 5.3.1 CAD Methodology and Model Conventions

The model is built top-down from the envelope closure of Section 5.2 rather than bottom-up from individual parts, so that the edge length $L$ and the wheel diameter $D_w$ propagate to every dependent feature when an iteration changes them. Three conventions make that propagation reliable, and they are stated here because they constrain how every part in Appendix B is modelled:

1. **A single skeleton drives the assembly.** $L$, $D_w$, the motor axis offsets and the wall thickness live in one parameter table referenced by every part; no downstream part carries a hard-coded dimension that duplicates one of them.
2. **The body frame is the CAD origin.** The assembly origin is the cube's geometric centre, with axes on the three motor spin axes, so that CAD-reported mass properties are directly the $I$ and $r_{cm}$ of Table 10 without a change of basis.

3. **Mass properties are an output of the model, not an annotation.** Every part carries its assigned material and print density, so the assembly mass roll-up is the quantity fed back into Table 13 at step 6 of Table 11.

Convention 3 is what makes the iteration auditable: the mass budget and the inertia requirement are read from the same model, so a design that closes on paper cannot silently fail to close in the CAD.

### 5.3.2 2D Prototype Build

The single-face rig is the first article built and the bench on which the parameters of Table 6 are measured. Its structure is a 3D-printed plate sized to one cube face, a central motor and wheel mount, a corner bearing defining the single rotational degree of freedom, and the servo brake. Detailed geometry and the drawing set are in Appendix B.3.

### 5.3.3 3D Cube Frame

The frame is conceived in two parts, reflecting the constraint chain of Section 5.2. An *inner structure* locates the three motors on mutually perpendicular axes through a common centre and carries the battery pack, the three drivers, the flight controller and the brake servos; it is the element that fixes the minimum internal clear volume and therefore the edge length $L$. An *outer frame* carries the edge and corner contact features, protects the wheels, and reacts the motor mounting loads from the inner structure into the ground contact. Detailed geometry is pending CAD and follows the envelope closure.

The split is also the main mechanical interface in the machine, and it is managed as one: the inner structure is designed against a fixed bolt pattern and keep-out envelope published to the outer frame, so the two can be revised independently as long as that interface holds. The controlled interfaces are listed in Table 14; the parts realising them, with their drawings, are in Appendix B.4.

**Table 14:** Controlled mechanical interfaces. Each is frozen once agreed, so that the parts on either side can be revised independently.

| Interface | Between | Controlled quantities |
|---|---|---|
| Motor mount | Motor ↔ inner structure | Bolt circle, shaft height, concentricity to the wheel axis |
| Wheel hub | Wheel ↔ motor rotor | Bore, bolt pattern, axial location, balance datum |
| Frame joint | Inner structure ↔ outer frame | Bolt pattern, keep-out envelope, load path to the contact features |
| Electronics mount | Board ↔ inner structure | Board outline, mounting-hole pattern, keep-out heights, connector exit directions (Sec. 5.4.5) |
| Brake mount | Servo ↔ inner structure | Servo location, barrier throw, clearance to the ballast bolt heads |
| Battery retention | Pack ↔ inner structure | Pack envelope, retention method, centring tolerance to the cube centre |
| Contact features | Outer frame ↔ ground | Corner and edge geometry, material, replaceability |

### 5.3.4 Reaction-Wheel Sizing

The wheel must supply the per-wheel inertia $I_{w,\text{target}}$ demanded by the corner jump-up (Section 4.3.1), at an outer diameter no greater than the envelope permits (step 3 of Table 11), while remaining light enough not to dominate the cube mass—since its own mass feeds back into $I_{w,\text{target}}$ through $M$. These three demands pull against each other, and the wheel architecture is chosen so that the design can be trimmed against them after fabrication rather than having to be right the first time.

**Architecture: ballasted ring.** Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel bolts and nuts seated in blind hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel at the largest available radius, so the net inertia gain per pocket is large, and the target is reached by choosing how many bolts to install. This makes wheel inertia a discrete, reversible knob: the physical wheel can be re-tuned in either direction after fabrication to absorb the parameter uncertainty that the sizing procedure necessarily carries—most importantly the residual uncertainty in $M$, which is not known until the build is complete. A solid disc offers no such adjustment.

The design intent is that the bare printed ring supplies most of the required inertia and the ballast acts only as a trim, so that the bolt count sits well inside its feasible range in both directions.

**Parameters to be fixed.** The geometry is defined by the quantities of Table 15, all of which follow from the envelope closure of Section 5.2 and are recorded there once fixed.

**Table 15:** Reaction-wheel design parameters, to be fixed once the envelope closes.

| Symbol | Parameter | Value |
|---|---|---|
| $D_w$ | Ring outer diameter (capped by envelope) | TBD |
| $w_r$ | Ring radial width | TBD |
| $t_r$ | Ring axial thickness | TBD |
| $n_s, w_s$ | Spoke count and width | TBD |
| $\rho_p$ | Printed-material density | TBD |
| $r_b$ | Ballast radius | TBD |
| $m_b$ | Net mass added per installed bolt | TBD |
| $\Delta I_b$ | Net inertia added per installed bolt | TBD |
| $N$ | Bolt count reaching $I_{w,\text{target}}$ | TBD |
| $m_w$ | Resulting wheel mass | TBD |
| $I_w$ | Resulting wheel inertia | TBD |

The ballast is characterised *net* of the plastic each pocket displaces: $m_b$ and $\Delta I_b$ are the increments the installed bolt-and-nut adds over the material it replaces, evaluated at the ballast radius. Sizing then reduces to solving

$$I_w(D_w, N) \;=\; I_{w,\text{ring}}(D_w) + N\,\Delta I_b \;\geq\; I_{w,\text{target}}$$

for the smallest admissible $N$ at each candidate $D_w$, subject to the fit and mass checks below.

**Trade study.** Table 16 is the record sheet for the outer-diameter sweep, populated once $I_{w,\text{target}}$ and the envelope cap are known. Each candidate diameter is evaluated on four criteria, and the selected design is the one that passes all of them with the widest trim margin:

1. **Envelope.** $D_w$ must clear the motor subframe, the motor body and the two orthogonal wheel modules inside the selected $L$.
2. **Inertia.** $I_w \geq I_{w,\text{target}}$, with margin for the uncertainty in $M$.
3. **Pocket fit.** Enough plastic wall must survive between adjacent pockets and radially inboard and outboard of each. Small diameters fail here: they need so many bolts to reach the target that no wall remains between them.
4. **Mass and trim margin.** The three-wheel set must stay within its allocation in Table 13. Diameters large enough that the bare ring already exceeds the target are rejected—they leave no downward trim authority and cost mass for nothing.

Criteria 3 and 4 bracket the feasible range from below and above respectively, so the selection is the largest diameter that still requires a non-trivial bolt count.

**Table 16:** Reaction-wheel outer-diameter trade study. To be populated once $I_{w,\text{target}}$ and the envelope cap are fixed; $N$ is the bolt count reaching the target, rounded up to a multiple of the spoke count for balance.

| $D_w$ (mm) | $N$ | Mass (g) | $I_w$ (kg m$^2$) | $I_w$/target (%) | 3 wheels (g) | Pocket fit |
|---|---|---|---|---|---|---|
| TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| TBD | TBD | TBD | TBD | TBD | TBD | TBD |

**Pocket placement.** The pockets cannot in general be spaced uniformly. A spoke occupies a finite angular width where it meets the ring, as does a pocket at the ballast radius, so a pocket centre must clear a spoke centre by the sum of the two half-widths. Whenever the uniform pitch $360°/N$ divides the spoke pitch exactly, a uniform arrangement places a pocket on every spoke root and is inadmissible. The pockets are therefore distributed in equal groups within each inter-spoke sector, spread over the usable arc that remains after the spoke exclusion zones are removed. Grouping by sector preserves the $n_s$-fold symmetry and hence static and dynamic balance, and the count per sector is chosen to leave spare positions for later trim.

**Bolt retention.** Each bolt carries a centrifugal load $F = m_b\,\omega_{\text{max}}^2\,r_b$ at the design speed, reacted by the pocket floor in bending. Positive retention is nonetheless required, and is provided by through-fastening each bolt with its nut against the pocket floor. This both retains the ballast and preserves the removability that motivates the scheme; the floor thickness and the pocket depth are set so that $F$ at the final $\omega_{\text{max}}$ is carried with margin.

### 5.4 Electrical Design

The electrical architecture is presented here at the level of the four decisions that shape it: what the components are, how power is distributed, how the buses are routed and terminated, and how the whole is made safe to power on. The sheet-level schematics, the connector and pin tables, the harness drawings and the perfboard layout are collected in Appendix C, for the same reason the CAD detail is separated from the mechanical rationale — a connector re-pin should not require an edit here.

### 5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A (Table 22). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team's design responsibility, and their fabrication is the leading schedule risk.

### 5.4.2 Power Architecture

The power architecture is shown in Figure 3. A single 6S LiPo pack (22.2 V nominal, 3600 mA h) feeds everything through one XT90 connector, behind a combined E-stop and fuse using an XT90-S anti-spark connector to limit inrush into the driver bulk capacitance. The bus then splits into two rails. The *motor bus* runs at pack voltage (22–25 V across the discharge curve) directly into the three moteus drivers, with bulk capacitance rated to at least 50 V to absorb regenerative braking transients—which are not incidental here, since the jump-up maneuver of Section 4.3 dumps the full wheel kinetic energy back into the bus. The *logic rail* is derived by an LM2596 buck converter stepping 22 V down to 5 V for the Teensy, the ESP32 telemetry link, the brake servo and the IMU. Keeping the two rails separate downstream of the E-stop means motor-current ripple does not reach the sensing and processing electronics.

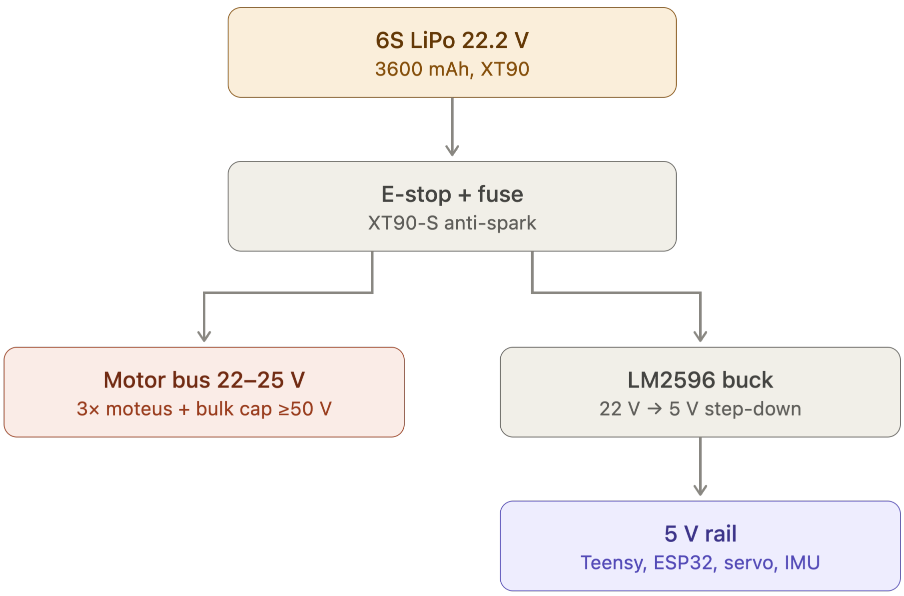


**Figure 3:** Power distribution tree. A single 6S pack feeds the motor bus directly and the 5 V logic rail through a buck converter, both behind a common E-stop and fuse.

The rail-by-rail current budget that sizes the buck converter and the harness conductors is Table 17. It is populated from measurement during bench bring-up rather than from datasheet maxima alone, because the figure that matters for the 5 V rail is the coincident worst case — telemetry transmit burst, servo stall and full sensor load at once — and not the sum of typical values.

**Table 17:** Rail current budget. Measured values replace estimates during bench bring-up; the 5 V total is checked against the LM2596 rating, which derates without a heatsink.

| Load | Rail | Typical (mA) | Worst case (mA) | Basis |
|---|---|---|---|---|
| Teensy 4.1 | 5 V | TBD | TBD | Datasheet / measured |
| XIAO ESP32-C6 telemetry | 5 V | TBD | TBD | TX burst governs |
| BMI270 IMU | 5 V | TBD | TBD | Datasheet |
| MG92B brake servo | 5 V | TBD | TBD | Stall current governs |
| MA600 encoders (3×) | 5 V | TBD | TBD | Datasheet |
| **Logic rail total** | 5 V | **TBD** | **TBD** | **vs. converter rating** |
| moteus-n1 drivers (3×) | Pack | TBD | TBD | Spin-up, not balance, governs |

### 5.4.3 Schematic and Interconnect Diagrams

The signal-level design is documented as a set of sheets rather than a single drawing, so that each can be checked against the constraint that governs it. Table 18 is the index; the sheets themselves are in Appendix C.1, together with the connector and pin-assignment tables that a person holding the harness needs and a reader of this section does not.

**Table 18:** Schematic sheet index. Each sheet is checked against the governing constraint stated in its row before release.

| Sheet | Content | Governing constraint to verify |
|---|---|---|
| 1 | Power entry and protection | E-stop, fuse rating, anti-spark inrush into the driver bulk capacitance |
| 2 | 5 V logic rail | Converter rating vs. Table 17 worst case; back-feed diode; rail bulk capacitance voltage rating |
| 3 | CAN-FD bus | Linear topology, stub length, split termination at the two physical ends only (Sec. 5.4.4) |
| 4 | Encoder interfaces (3×) | SPI cable length and encoder air gap; driver placement follows from this, not the reverse |
| 5 | Flight computer and IMU | Sensor orientation and mounting location relative to the pivot (Sec. 4.4) |
| 6 | Telemetry link | Transmit-burst current and its return path off the logic rail |
| 7 | Brake servo drive | Stall current headroom on the logic rail; PWM routing away from the encoder SPI |

### 5.4.4 Bus Topology, Termination and Signal Integrity

The CAN-FD bus is the one electrical subsystem whose physical routing is a correctness requirement rather than a packaging preference, and it is therefore fixed before the board outline is drawn. At the design data rate the bus must be a linear chain with short stubs and termination at exactly the two physical ends; a star topology or a mid-chain termination will pass a bench continuity check and then fail intermittently under wheel vibration, which is the hardest class of fault to diagnose once the cube is closed. The design rules and their verification are listed in Table 19.

The encoder interfaces impose the complementary constraint. Each absolute encoder requires a short SPI run to its own driver and a tightly held air gap to its magnet, so the driver positions are dictated by the encoder positions, and those in turn by the motor axes. The physical layout order is therefore: motor axes fix the encoders, encoders fix the drivers, drivers fix the bus route, and only then is the board outline of Section 5.4.5 drawn around what remains.

**Table 19:** Signal-integrity design rules and how each is verified. These are checked before cube closure, since all are hard to access afterwards.

| Interface | Rule | Verification |
|---|---|---|
| CAN-FD bus | Linear chain; short stubs; split termination at the two physical ends only | Differential-pair scope capture; sustained error-frame count at full rate with all nodes addressed |
| Encoder SPI | Short run to its own driver; air gap and chip centring per the assembly spec | Read-back with no error flags; re-checked after mechanical assembly |
| IMU | Mounted at or near the pivot; orientation per the body frame of Sec. 5.3.1 | Static orientation check; noise floor with wheels spinning |
| Rail separation | Motor bus and logic rail separated downstream of the E-stop | Ripple measurement on the logic rail under motor load |
| Grounding | Single return topology; no motor return current through the sensing ground | Continuity and isolation check before first power-on |

### 5.4.5 Board Layout and Harness

The electronics are built on a double-sided perfboard rather than a fabricated PCB. This is a schedule decision, not a technical preference: fabrication turnaround does not fit inside the build window, and a perfboard can be modified in place when bring-up reveals a change. Its consequence is that layout discipline — rail separation, decoupling at each device, deliberate routing of the bus pair — must be enforced by hand and checked explicitly, since no design-rule check will do it.

The board outline, mounting-hole pattern, keep-out heights and connector exit directions constitute the electronics-mount interface of Table 14 and are frozen once delivered to the mechanical design: later electrical changes must fit inside that outline rather than move it. The harness — battery lead, driver drops, bus chain and servo drive — is fabricated to the drawings in Appendix C.3 and is labelled and strain-relieved before installation, because every connector in the machine sits on a structure that vibrates by design.

### 5.4.6 Electrical Safety and Bring-Up Protocol

Two rules govern all electrical bring-up and are stated here because they are design constraints, not merely lab practice. First, every first-time power application is made on a current-limited bench supply, never on the battery; the battery is introduced only after the assembly has passed a full bench power-on. Second, the pack is never connected without the E-stop and fuse in circuit and the anti-spark path intact, since the driver bulk capacitance draws an inrush that will weld a bare connector.

The staged sequence — rails, then buses, then sensors, then open-loop motion, then closed-loop — is part of the validation plan and is given in Section 6. The isolation checks that precede first power-on (rail-to-rail separation, bus-to-chassis, and confirmation that no low-voltage capacitor sits on the pack bus) are recorded in Appendix C as a checklist, because they are performed on every re-assembly and not only once.

## 5.5 System Integration

### 5.5.1 Onboard Software Pipeline

The subsystems of Figure 2 are coordinated by a single deterministic task on the Teensy, whose structure is given in Figure 4. Each 1 ms cycle executes the same fixed sequence: sensor ingest reads the BMI270 over SPI and the moteus replies over CAN; the MEKF propagates attitude on the gyro and corrects it against the accelerometer while estimating gyro bias; the state vector $x = [g_B, \omega, h]$ is assembled from the estimator output and the wheel speeds; the LQR law of Section 4.4 evaluates $u = -Kx$ with a penalty on stored momentum $h$ to bleed off wheel saturation; and the resulting commands leave as CAN torque requests to the drivers and a PWM signal to the brake servo. Telemetry is tapped off the assembled state to SD and Wi-Fi outside the control path, so logging cannot perturb loop timing.

The fixed ordering matters: with no dynamic allocation and no blocking calls, the worst-case cycle time is bounded by construction rather than measured after the fact, which is what makes the 1 kHz sample period of the discrete LQR design a design guarantee rather than an average.

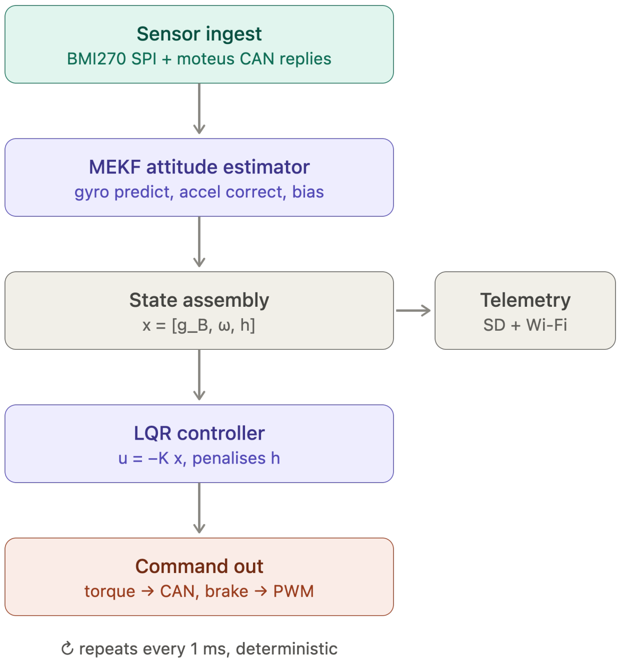


**Figure 4:** Teensy 4.1 control-loop software pipeline. The estimator and controller stages implement the design of Section 4.4; telemetry is a side-branch off the control path.

### 5.5.2 Control Software Architecture

Section 4 derives the control law; this subsection states how that law is realised as software, which is a separate set of decisions. The implementation detail — the mode state machine, the estimator and controller in executable pseudocode, the fixed gains and tuning parameters, the timing budget and the fault responses — is collected in Appendix D, so that a re-tuned gain does not require an edit to the design sections.

The software is organised in the four layers of Table 20. The division is not stylistic: each layer has a different failure mode and a different verification method, and the interfaces between them are what allow the controller validated in simulation to be the same code that runs on the cube.

**Table 20:** Control software layers. Each is verified by a different method, which is why the boundaries are held.

| Layer | Responsibility | Verified by |
|---|---|---|
| Hardware abstraction | Sensor and driver I/O, timing, bus transactions | Bench read-back; loop-timing measurement |
| Estimation | Attitude and rate estimate from IMU and encoders (Sec. 4.4) | Static and dynamic bench tests against a known reference |
| Control | State feedback, momentum management, saturation handling | Simulation on the identified model, then hardware |
| Supervision | Mode transitions, arming, watchdog and fault response | Fault-injection tests on the rig |

**Operating modes.** The machine is never in a single control regime. The supervisory layer sequences the modes of Table 21, with hysteresis on every threshold to prevent chattering at a boundary, and with a disarm transition available from every mode. Documenting the modes as a state machine rather than as a set of conditionals matters here because the dangerous transitions are the ones back down the ladder: a fall detected during corner balance must reach a safe state with the wheels commanded down, not merely switch to a different gain.

**Table 21:** Operating modes and transitions. Every mode has a disarm exit; the detailed conditions and hysteresis bands are in Appendix D.

| Mode | Behaviour | Exit condition |
|---|---|---|
| Idle / disarmed | Drivers de-energised; sensors and telemetry live | Explicit arm command |
| Wheel spin-up | Open-loop speed ramp to the jump-up momentum target | Target wheel speed reached |
| Jump-up | Brake actuation and momentum transfer (Sec. 4.3) | Tilt inside the balancing capture band |
| Edge balance | Single-axis state feedback (Sec. 4.4) | Corner-transition command, or fall detected |
| Corner balance | Three-axis state feedback (Sec. 4.5) | Slew command, or fall detected |
| Slew / rotation | Reference tracking between equilibria | Reference reached; revert to balance |
| Fault / safe | Wheels commanded down, drivers disarmed | Manual reset |

**Real-time guarantees.** The loop-rate claim of Section 4.4 is a design guarantee only if the worst-case cycle is bounded by construction, which is why the pipeline of Figure 4 uses a fixed stage ordering with no dynamic allocation and no blocking calls, and why telemetry is a side-branch rather than a stage. The per-stage timing budget and its measured margin are recorded in Appendix D and re-measured whenever a stage changes.

**Fault handling.** The failure cases the software must survive are sensor dropout, bus error, wheel saturation and loss of the balancing equilibrium. The first three have defined responses that keep the machine controlled; the fourth is a controlled shutdown rather than a recovery attempt, since a cube that has left its capture region cannot be recovered by the linear law and continued actuation only adds energy to the fall.

# 6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section 4.2.

## 6.1 Simulation-First Validation

## 6.2 Bring-up Sequence

## 6.3 Sensor Testing

## 6.4 Milestone Acceptance Criteria

## 6.5 Control Performance Evaluation

# 7 Future Work

## 7.1 NMPC & Warm-Start Control (Future Implementation)

## 7.2 Autonomous Stand-up & Disturbance Rejection

## 7.3 Single-Wheel Variant / Extensions

## References


[1] Mohanarajah Gajamohan, Michael Merz, Igor Thommen, and Raffaello D'Andrea. "The Cubli: A Cube that can Jump Up and Balance". In: *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Vilamoura-Algarve, Portugal, Oct. 2012, pp. 3722–3727. DOI: 10.1109/IROS.2012.6385896.

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# A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team's design responsibility, and are treated as such in Section 5. Table 22 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 4.3.1 and 5.3.4.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build, and with three actuation sets and no spare, the loss of a motor or driver in service would force the reduced-actuation one-wheel variant [8]. These items are ordered first. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, filament, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation.

**Table 22:** Supplied bill of materials, grouped by subsystem.

| Ref | Item | Qty[a] | Key specifications | Function |
|---|---|---|---|---|
| **Actuation** | | | | |
| 1 | T-Motor MN4006 KV380 | 3 | 18N24P outrunner; 380 rpm/V; 68 g; ⌀44.35 mm × 21 mm; 194 mΩ; peak 16 A, 380 W | Reaction-wheel torque source |
| 2 | mjbots moteus-n1 | 3 | 10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN-FD; 14.6 g | FOC current/torque driver (inner loop) |
| 11 | TowerPro MG92B servo | 1–3 | metal-gear micro servo; ≈0.29 N m (≈3 kg cm) at 6 V; 50 Hz PWM | Wheel brake actuator (jump-up) |
| **Sensing** | | | | |
| 3 | mjbots MA600 breakout + ring magnet | 3 | TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm | Wheel angle: commutation + state feedback |
| 10 | SparkFun BMI270 6-DoF IMU | 1 | 3-axis accel + 3-axis gyro (no magnetometer); $I^2C$ 400 kHz (Qwiic); ODR up to ≈ 1 kHz | Body attitude reference (tilt + rates) |
| **Compute and Communications** | | | | |
| 4 | Teensy 4.1 | 1 | 600 MHz Cortex-M7 + FPU; native CAN-FD (CAN3); 5 V supply | Flight computer: estimator + control law |
| 5 | CAN-FD adapter (Teensy 4.1) | 1 | CAN-FD transceiver, logic-level → differential; 5 Mbit/s | Bus transceiver for the MCU |

*Continued on next page*

*Table 22 (continued)*

| Ref | Item | Qty[a] | Key specifications | Function |
|---|---|---|---|---|
| 6 | mjbots mjcanfd-usb-1x | 1 | USB ↔ CAN-FD, 5 Mbit/s | Bench interface (calibration, tuning) |
| 7 | JST-PH3 CAN cables | ≥4 | TODO: conductor gauge / current rating not in source; 3-conductor daisy-chain | CAN bus interconnect |
| 8 | Seeed XIAO ESP32-C6 + antenna | 1 | Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA | Wi-Fi telemetry / command link |
| 9 | 60.4 Ω 0.5 W resistor | 4 | split termination, $2 \times 60.4\,\Omega \approx 120\,\Omega$ bus impedance | CAN bus termination |
| 17 | 4.7 nF capacitor | 2 | common-mode filter, bus midpoint to ground | CAN split-termination filter |
| **Power** | | | | |
| 12 | Turnigy 6S LiPo (XT90) | 1 | 22.2 V nom / 25.2 V charged; 3600 mA h; 60C; ≈ 80 W h; ≈ 600 g | Primary energy store |
| 13 | LM2596 buck module | 1 | step-down to 5.0 V; rated 2–3 A (derates without heatsink) | 5 V logic rail |
| 14 | XT90 connector pair | 1–2 | rated ≈ 90 A | Battery power connector |
| 15 | 100 µF 16 V electrolytic | 2–4 | 16 V part — 5 V rail only | 5 V-rail bulk reservoir |
| 19 | 1N5819 Schottky diode | 1–2 | 1 A, 40 V | 5 V-rail back-feed protection |
| **Integration** | | | | |
| 16 | 100 nF 50 V capacitor | 6–10 | local decoupling at each IC | Per-IC supply decoupling |
| 18 | Double-sided perfboard | 1 | 15 cm × 9 cm, double-sided | Power / signal distribution board |

[a] Quantity per assembled cube; motors, drivers and encoders are 3 each. A fourth actuation set (motor + moteus-n1 + MA600) is retained on the 1-DoF bench rig (S1) throughout the build, so four of each are procured.

# B CAD and Construction Detail

This appendix holds the mechanical detail deliberately kept out of Section 5.3: the model tree, the drawing set, the fabrication settings and the assembly sequence. The rationale for the architecture — why the frame is split, what fixes the envelope, how the wheel is tuned — stays in Section 5.3 and is not repeated here. Anything in this appendix may change without the design intent changing; anything in Section 5.3 may not.

## B.1 Design Parameter Table

Table 23 is the single source of the driving dimensions referenced by every part in the model, per convention 1 of Section 5.3.1. It is the CAD-side mirror of the envelope closure of Section 5.2 and is updated on each iteration.

**Table 23:** Driving CAD parameters. Every dependent feature references these; no part carries a duplicated hard-coded value.

| Symbol | Parameter | Value |
|---|---|---|
| $L$ | Cube edge length (outer) | TBD |
| $t_w$ | Frame wall thickness | TBD |
| $D_w$ | Wheel outer diameter (Sec. 5.3.4) | TBD |
| $d_m$ | Motor axis offset from cube centre | TBD |
| $h_m$ | Motor mounting face height on its axis | TBD |
| $c_w$ | Wheel-to-subframe axial clearance | TBD |
| $c_o$ | Clearance between orthogonal wheel modules | TBD |
| $V_{\min}$ | Minimum internal clear volume | TBD |
| $r_c$ | Corner contact feature radius | TBD |

## B.2 Model Tree and Part List

Table 24 lists the modelled parts by assembly, with their fabrication route and current status. It is the mechanical counterpart to the electronics bill of materials of Appendix A: the parts here are the team's design responsibility and are not supplied.

## B.3 2D Prototype Construction

## B.4 3D Cube Construction

## B.5 Fabrication and Assembly

### B.5.1 Print and Manufacturing Settings

The printed parts are structural, not cosmetic: the wheel ring carries centrifugal load at the ballast pockets and the motor mounts carry the brake impulse, which is the design load case identified in Section 1.1. Settings are therefore recorded per part in Table 25 and treated as part of the design, since a part reprinted at a different orientation or infill is not the part that was analysed.

### B.5.2 Assembly Sequence

The sequence of Table 26 is ordered by access: each step is one whose verification becomes impossible or destructive once a later step is complete. The encoder air gap in particular is rechecked after mechanical assembly, since it can shift when the structure is bolted up (Table 19).

**Table 24:** CAD model tree and part list. Quantities are per assembled cube unless noted; the 1-DoF rig parts are listed separately.

| Ref | Part | Qty | Fabrication route | Status |
|---|---|---|---|---|
| **Inner structure** | | | | |
| M1 | Motor subframe | TBD | TBD | TBD |
| M2 | Motor mount (×axis) | 3 | TBD | TBD |
| M3 | Battery cradle / retention | 1 | TBD | TBD |
| M4 | Electronics tray | 1 | TBD | TBD |
| M5 | Brake servo bracket | 3 | TBD | TBD |
| **Reaction wheel assembly** | | | | |
| W1 | Ballasted ring (Sec. 5.3.4) | 3 | TBD | TBD |
| W2 | Wheel hub / rotor adapter | 3 | TBD | TBD |
| W3 | Ballast bolt and nut set | TBD | Standard hardware | TBD |
| W4 | Brake strike feature | 3 | TBD | TBD |
| **Outer frame** | | | | |
| F1 | Frame member / edge rail | TBD | TBD | TBD |
| F2 | Corner contact feature | 8 | TBD | TBD |
| F3 | Face panel / wheel guard | TBD | TBD | TBD |
| **1-DoF rig (Sec. 5.3.2)** | | | | |
| R1 | Face plate | 1 | TBD | TBD |
| R2 | Pivot bearing mount | 1 | TBD | TBD |
| R3 | Rig motor mount | 1 | TBD | TBD |
| R4 | Hard stops / containment | TBD | TBD | TBD |

## B.6 Mass Properties Verification

The CAD roll-up is an estimate until the assembly is weighed. Step 8 above closes the loop of Section 5.2: the measured mass is compared against the modelled value, and any discrepancy is carried back into $I_{w,\mathrm{target}}$ and absorbed, where possible, by the ballast trim of Section 5.3.4 rather than by a re-design.

**Table 25:** Fabrication settings for the structural printed parts. Layer orientation is specified relative to the principal load direction.

| Ref | Material | Infill / walls | Orientation and rationale |
|---|---|---|---|
| W1 | TBD | TBD | TBD |
| W2 | TBD | TBD | TBD |
| M1 | TBD | TBD | TBD |
| M2 | TBD | TBD | TBD |
| F1 | TBD | TBD | TBD |
| F2 | TBD | TBD | TBD |

**Table 26:** Assembly sequence. Ordering is set by access: each check must be completed before the step that encloses it.

| Step | Operation | Check before proceeding |
|---|---|---|
| 1 | Wheel assembly: ring, hub, ballast bolts to the selected count | Balance check; bolt retention per Sec. 5.3.4 |
| 2 | Motor to mount, encoder magnet to rotor | Encoder air gap and centring to the assembly spec |
| 3 | Motor modules to inner structure, three axes | Axis orthogonality; wheel clearance $c_w$, $c_o$ |
| 4 | Battery and electronics tray into inner structure | Pack centring; board keep-out clearances |
| 5 | Harness install and dressing | Continuity and isolation (Sec. 5.4.6) |
| 6 | Inner structure into outer frame | Encoder air gaps re-checked after bolt-up |
| 7 | Contact features and guards | Contact geometry; wheel guard clearance |
| 8 | Mass properties measurement | Measured mass against Table 13 |

**Table 27:** Modelled versus measured mass properties. A discrepancy here is absorbed by ballast trim where the margin allows.

| Quantity | CAD value | Measured | Discrepancy |
|---|---|---|---|
| Total cube mass $M$ | TBD | TBD | TBD |
| Wheel mass $m_w$ (each) | TBD | TBD | TBD |
| Wheel inertia $I_w$ (each) | TBD | TBD | TBD |
| CoM offset from cube centre | TBD | TBD | TBD |

# C Electrical System Detail

This appendix holds the electrical detail kept out of Section 5.4: the schematic sheets indexed in Table 18, the connector and pin assignments, the harness drawings, the board layout and the bring-up checklists. The architectural decisions — two rails behind one E-stop, linear bus topology, perfboard rather than fabricated board — remain in Section 5.4. The part specifications are in Appendix A and are not repeated.

## C.1 Schematic Sheets

### C.1.1 Sheet 1 — Power Entry and Protection

### C.1.2 Sheet 2 — Logic Rail

### C.1.3 Sheet 3 — CAN-FD Bus

### C.1.4 Sheet 4 — Encoder Interfaces

### C.1.5 Sheet 5 — Flight Computer and IMU

### C.1.6 Sheet 6 — Telemetry Link

### C.1.7 Sheet 7 — Brake Servo Drive

## C.2 Connector and Pin Assignments

Table 28 is the connector schedule. It exists so that a harness can be built and re-built from this document alone, and so that a mis-mate is a documented impossibility rather than a discovered one: no two connectors carrying different voltages share a housing type on this machine.

**Table 28:** Connector schedule. Keying and housing types are chosen so that two connectors carrying different voltages cannot be mated.

| Ref | From | To | Housing / keying | Signals |
|---|---|---|---|---|
| J1 | Battery pack | Protection / E-stop | TBD | Pack +, pack − |
| J2 | Protection | Distribution board | TBD | Motor bus |
| J3 | Distribution board | Driver, axis 1 | TBD | Motor bus |
| J4 | Distribution board | Driver, axis 2 | TBD | Motor bus |
| J5 | Distribution board | Driver, axis 3 | TBD | Motor bus |
| J6 | Board | CAN chain, end A | TBD | CAN-H, CAN-L, GND |
| J7 | CAN chain, end B | Termination network | TBD | CAN-H, CAN-L, GND |
| J8 | Driver, axis $i$ | Encoder, axis $i$ | TBD | SPI, 3.3 V, GND |
| J9 | Board | IMU | TBD | Serial bus, supply, GND |
| J10 | Board | Telemetry module | TBD | UART, supply, GND |
| J11 | Board | Brake servo, axis $i$ | TBD | PWM, 5 V, GND |

## C.3 Harness Drawings and Wire Schedule

Table 29 is the wire schedule. Conductor gauge on the motor bus is set by the spin-up current rather than the balancing current, since spin-up is the sustained high-current phase; the signal runs are set by length limits rather than by current.

**Table 29:** Wire schedule. Motor-bus gauge is sized on spin-up current; signal runs are length-limited (Table 19).

| Ref | Run | Gauge | Length | Sizing basis |
|---|---|---|---|---|
| H1 | Pack to protection | TBD | TBD | Peak bus current |
| H2 | Protection to board | TBD | TBD | Peak bus current |
| H3 | Board to driver drops (3) | TBD | TBD | Per-axis spin-up current |
| H4 | Motor phase leads (3) | TBD | TBD | Motor phase current |
| H5 | CAN chain segments | TBD | TBD | Topology, not current |
| H6 | Encoder SPI runs (3) | TBD | TBD | Length limit governs |
| H7 | Servo drives (3) | TBD | TBD | Stall current |

## C.4 Board Layout

The floorplan is constrained before it is aesthetic: the driver positions follow from the encoder cable limit, the bus route follows from the driver positions, and the board is placed around what remains (Section 5.4.4). Rail separation is maintained physically on the board, not only schematically, so that motor-current return does not share copper with the sensing ground.

## C.5 Bring-Up and Isolation Checklists

These checks are performed on every re-assembly, not only on first build, since the encoder gaps and connector seating are exactly what mechanical work disturbs.

**Before any power is applied.**

1. Isolation: motor bus to chassis, logic rail to motor bus, each rail to ground.
2. Capacitor voltage ratings confirmed by position — no low-voltage part on the pack bus.
3. Polarity of every fabricated connector verified against Table 28.
4. Protection element in circuit: E-stop functional, fuse fitted and rated, anti-spark path intact.
5. Encoder air gaps re-measured after mechanical assembly.

**First power-on, current-limited bench supply.**

1. Current limit verified against a known load before connecting the assembly.
2. Logic rail voltage and ripple under load; converter thermal check.
3. All drivers enumerate on the bus; encoders read back with no error flags; IMU responds.
4. Bus integrity at the full data rate: sustained soak with all nodes addressed, error-frame count logged.
5. Open-loop wheel motion, wheels contained, one axis at a time.

**First battery power-on.** Performed only after the bench sequence passes in full, with the pack protection verified, cell monitoring active and the assembly in a fire-safe area. Inrush is measured against the bulk capacitance on connection.

## C.6 As-Built Measurements

Table 30 records the measured electrical figures that replace the estimates used during design. These are the numbers reported back to the modelling and control work, and they are the reason the parameter identification of Section 4.2 runs on measured rather than assumed values.

**Table 30:** As-built electrical measurements. Each replaces an estimate used earlier in the design.

| Quantity | Measured | Replaces / feeds |
|---|---|---|
| Motor torque constant $K_m$ | TBD | Table 6, datasheet value |
| Wheel inertia $I_w$ (spin-down) | TBD | Table 6, CAD value |
| Viscous and Coulomb friction | TBD | $C_b$, $C_w$ estimates |
| Command-to-encoder latency | TBD | Loop timing, Sec. 5.5.2 |
| Achievable wheel speed on bus | TBD | $\omega_{\max}$, Sec. 4.3.1 |
| Logic rail load, worst case | TBD | Table 17 |
| Runtime per charge under balance load | TBD | Demonstration planning |

# D Control Algorithms

This appendix holds the implementation detail of the control software whose architecture is given in Section 5.5.2. The derivations remain in Section 4 and are not repeated; what follows is the executable form — the loop, the estimator, the control law, the mode logic and the fault responses — together with the gains and timing figures that are re-tuned against hardware and therefore change more often than the design does.

## D.1 Control Loop

The top-level loop is the fixed sequence of Figure 4, executed at the sample period $\Delta t$ of Table 9. Its ordering is a real-time requirement, not a convenience: with no dynamic allocation and no blocking call, the worst-case cycle is bounded by construction.

```
every dt:
    t0 <- now()

    # 1. Sensor ingest -- fixed-size, non-blocking reads only
    imu      <- read_imu()                # rates and specific force
    wheels   <- read_driver_replies()     # wheel angle and speed, per axis
    flag_stale(imu, wheels)               # feeds the fault logic below

    # 2. Estimation -- propagate on rates, correct on the absolute reference
    xhat     <- estimator_predict(xhat, imu.rates, dt)
    xhat     <- estimator_correct(xhat, imu.accel)

    # 3. State assembly
    x        <- assemble_state(xhat, wheels)

    # 4. Supervision -- mode first, so the control law below is unambiguous
    mode     <- supervisor_update(mode, x, faults)

    # 5. Control law for the current mode
    u        <- control_law(mode, x, gains[mode])
    u        <- apply_limits(u)           # torque and momentum saturation

    # 6. Output
    send_torque_commands(u)
    set_brake(mode)

    # 7. Telemetry -- side-branch, never in the control path
    telemetry_publish_nonblocking(x, u, mode)

    log_cycle_time(now() - t0)            # feeds the timing budget
```

Telemetry sits after the output stage and is non-blocking by construction, so that logging cannot perturb loop timing — the property that makes the sample period a guarantee rather than an average.

## D.2 Attitude Estimation

The estimator propagates attitude on the measured rates and corrects it against the absolute reference, while estimating the rate-sensor bias. The two error sources it must handle pull in opposite directions and are the reason a single sensor will not do: the rate measurement is clean over a cycle but its bias drifts, and the absolute reference is drift-free but is corrupted by the vibration the machine itself generates by spinning the wheels (Section 1.1).

```
estimator_predict(xhat, rates, dt):
    w_corrected <- rates - xhat.bias
    xhat.att    <- integrate_attitude(xhat.att, w_corrected, dt)
    xhat.P      <- propagate_covariance(xhat.P, dt)   # bias random walk
    return xhat

estimator_correct(xhat, accel):
    # Gate the correction: during high-acceleration phases the absolute
    # reference is not measuring the vertical, and applying it would inject
    # the manoeuvre into the attitude estimate.
    if not accel_trustworthy(accel):
        return xhat
    innovation <- reference_direction(accel) - predicted_direction(xhat.att)
    K          <- gain(xhat.P, R_accel)
    xhat       <- apply_correction(xhat, K, innovation)   # attitude and bias
    return xhat
```

The gating in `estimator_correct` is the load-bearing part. The jump-up and any aggressive recovery are exactly the phases in which the accelerometer does not measure the vertical, so an ungated correction would drive the estimate with the manoeuvre. The trust test and its threshold are recorded in Table 33.

## D.3 Control Law

The balancing law is the state feedback of Sections 4.4 and 4.5, with two additions required by hardware: the stored-momentum penalty that bleeds off wheel speed, and the offset filter that absorbs a constant tilt-sensor bias without a calibration stage.

```
control_law(mode, x, K):
    if mode in {EDGE_BALANCE, CORNER_BALANCE, SLEW}:
        e       <- x.att - reference(mode)   # zero for pure regulation
        x_f     <- lowpass(x_f, e)           # absorbs constant sensor offset
        u       <- -K @ [e - x_f, x.rates, x.wheel_speeds]
        return u

    if mode == SPINUP:
        return speed_ramp(x.wheel_speeds, target_speed)

    if mode == JUMPUP:
        return open_loop_jump_sequence(t_since_mode_entry)

    return zero_torque()                     # IDLE and FAULT
```

The wheel-speed term in the feedback is not merely a damping state. Under any persistent disturbance the wheels drift toward their speed limit, so the gain on stored momentum is what keeps the actuators inside their envelope; without it the machine balances correctly until it saturates and then does not.

**Saturation handling.** `apply_limits` enforces the torque limit and the momentum envelope. Torque is clipped per axis; momentum saturation is handled by the speed penalty above rather than by clipping, since clipping a saturated wheel removes control authority precisely when it is needed. When a wheel reaches its limit despite the penalty, the supervisor is notified and treats it as the fault of Section D.5.

## D.4 Mode Supervision

The supervisor implements the transitions of Table 21. Every threshold carries hysteresis, and every mode has a disarm exit.

```
supervisor_update(mode, x, faults):
    if faults.critical:
        return FAULT                      # from any mode, unconditionally

    switch mode:
      IDLE:            if arm_command:              -> SPINUP
      SPINUP:          if wheel_speed >= target:    -> JUMPUP
      JUMPUP:          if |tilt| < capture_band:    -> EDGE_BALANCE
                       if timeout:                  -> FAULT
      EDGE_BALANCE:    if |tilt| > release_band:    -> FAULT      # fall detected
                       if corner_command:           -> CORNER_BALANCE
      CORNER_BALANCE:  if |tilt| > release_band:    -> FAULT
                       if slew_command:             -> SLEW
      SLEW:            if reference_reached:        -> CORNER_BALANCE
                       if |tilt| > release_band:    -> FAULT
      FAULT:           if manual_reset and safe:    -> IDLE
    return mode
```

The capture and release bands differ deliberately — entry into balancing is stricter than the condition for staying in it — which is the hysteresis that prevents chattering at a mode boundary.

## D.5 Fault Detection and Response

Table 31 gives the detection test and response for each failure case. The distinction that matters is between faults the machine can continue through and faults it cannot: only the loss of the balancing equilibrium terminates the run, because a body outside its capture region cannot be recovered by the linear law, and continued actuation adds energy to a fall that is already happening.

## D.6 Timing Budget

Table 32 records the per-stage timing budget and the measured margin. It is re-measured whenever a stage changes, because the guarantee of Section 5.5.2 is a claim about the worst case and not about the mean.

## D.7 Gains and Tuning Parameters

Table 33 is the tuning record: every constant the control software carries, with the source that fixes it. Values computed from the model are distinguished from values tuned against hardware, since only the latter must be re-established when the machine is re-built.

**Table 31:** Fault detection and response. Only loss of equilibrium terminates the run; the remaining cases keep the machine controlled.

| Fault | Detection | Response |
|---|---|---|
| Sensor dropout / stale data | Missing or repeated sample within the cycle deadline | Propagate on the remaining sensor; escalate to FAULT if persistent beyond the tolerance |
| Bus error | Error-frame count or missed driver reply | Retry within the cycle; escalate on repetition |
| Encoder error flag | Flag set, or angle discontinuity beyond the physical rate limit | Disarm the affected axis; FAULT if it is load-bearing for the current mode |
| Wheel saturation | Speed at limit despite the momentum penalty | Notify supervisor; reduce reference authority; FAULT if sustained |
| Loss of equilibrium | Tilt outside the release band | FAULT: wheels commanded down, drivers disarmed |
| Loop overrun | Measured cycle time exceeds the deadline | Log; FAULT if repeated, since the timing guarantee no longer holds |
| Watchdog | Loop fails to service the timer | Hardware disarm, independent of the software path |

**Table 32:** Per-cycle timing budget against the sample period. Re-measured whenever a stage changes.

| Stage | Budget | Measured worst case |
|---|---|---|
| Sensor ingest | TBD | TBD |
| Estimation | TBD | TBD |
| State assembly | TBD | TBD |
| Supervision | TBD | TBD |
| Control law | TBD | TBD |
| Command output | TBD | TBD |
| **Total cycle** | **TBD** | **TBD** |
| **Margin to deadline** | — | **TBD** |

**Table 33:** Control and estimation parameters. "Model" values follow from the design; "bench" values are tuned against hardware and re-established after a re-build.

| Symbol | Parameter | Source | Value |
|---|---|---|---|
| $\Delta t$ | Sample period | Model | TBD |
| $Q, R$ | State and input weights | Model | TBD |
| $K_d$ (edge) | Feedback gain, edge balance | Model | TBD |
| $K_d$ (corner) | Feedback gain, corner balance | Model | TBD |
| — | Momentum-penalty weight | Bench | TBD |
| $\gamma$ | Estimator correction weight | Bench | TBD |
| — | Accelerometer trust threshold | Bench | TBD |
| $\alpha$ | Offset-filter coefficient | Bench | TBD |
| $\theta_{\text{th}}$ | Balancing capture band | Bench | TBD |
| — | Balancing release band (hysteresis) | Bench | TBD |
| — | Spin-up target wheel speed | Model | TBD |
| — | Brake actuation timing | Bench | TBD |
| — | Torque limit per axis | Model | TBD |
| — | Watchdog timeout | Model | TBD |